

British Math Olympiad 2019 Round 1 — P5/6

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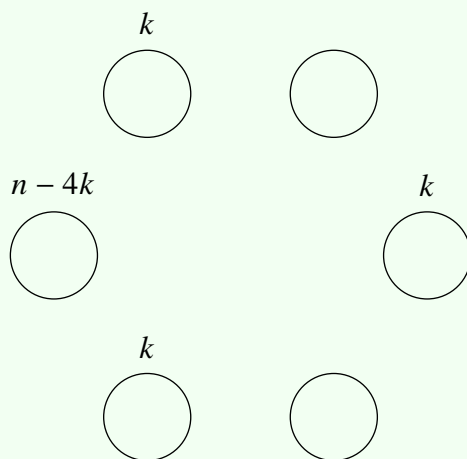
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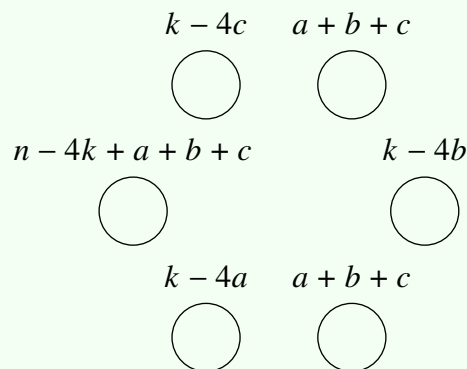
Six children are evenly spaced around a circular table. Initially, one has a pile of $n > 0$ sweets in front of them, and the others have nothing. If a child has at least four sweets in front of them, they may perform the following move: eat one sweet and give one sweet to each of their immediate neighbours and to the child directly opposite them. An arrangement is called perfect if there is a sequence of moves which results in each child having the same number of sweets in front of them. For which values of n is the initial arrangement perfect?

Solution

Suppose that, without loss of generality, the leftmost child starts off with the n sweets. Now suppose that the child performs the given operation a total of k times, the distribution of sweets will look like so,



Now suppose each of the children who currently have k sweets now perform the given operation a , b and c times respectively. We end up with the following situation,



Now if the two children who have $a + b + c$ sweets perform the given operation then the number of sweets the leftmost child has does not change. Also the number of sweets the children with $k - 4x$ sweets will gain is the same for all of them. Thus we must have $k - 4a = k - 4b = k - 4c$ and so $a = b = c$. Also $n - 4k + 3a = k - 4a + y$, where y is how many sweets the children with $k - 4x$ sweets will gain.